S.6 MATHEMATICS SEMINAR HELD ON 19th/SEPT/2026.

ADVANCED LEVEL PAPER STRUCTURE FOR P425/1 AND P425/2

PAPER 1 (P425/1)	PAPER 2 (P425/2)
1. ALGEBRA 2. GEOMETRY 3. CALCULUS	1. NUMERICAL METHODS 2. DATA ANALYSIS & PROBABILITY 3. MECHANICS
CONSTRUCT 1: GEOMETRY: item1 1. COORDINATE GEOMETRY 1 2. TRIGONOMETRY 3. VECTORS 4. COORDINATE GEOMTRY 2	**CONSTRUCT 1: NUMERICAL METHODS: item1** 1. ERROR ANALYSIS 2. TRAPEZIUM RULE 3. ITERATIVE METHODS 4. FLOW CHARTS
CONSTRUCT 2: ALGEBRA: items 2 & 3 1. NUMERICAL CONCEPTS 2. EQUATIONS AND INEQUALITIES 3. PERMUTATIONS AND COMBINATIONS 4. BINOMIAL THEOREM 5. SERIES 6. COMPLEX NUMBERS	**CONSTRUCT 2:DATA ANALYSIS &PROBABILITY: items 2 & 3** 1. DESCRIPTIVE STATISTICS 2. CORRELATION AND SCATTER DIAGRAMS 3. PROBABILITY THEORY 4. RANDOM VARIABLES 5. PROBABILITY DISTRIBUTIONS 6. SAMPLING DISTRIBUTIONS
CONSTRUCT 3: CALCULUS: items 4&5 1. PARTIAL FRACTIONS 2. DIFFERENTIATION 1 3. INTEGRATION 1 4. DIFFERENTIATION 2 5. INTEGRATION 2 6. DIFFERENTIAL EQUATIONS	**CONSTRUCT 3: MECHANICS: items 4 & 5** 1. DYNAMICS 1 2. DYNAMICS 2

PRINCIPAL MATHS ONE SET 1

ITEM ONE

Mr. Patu, a seasoned businessman intends to install solar panels on his newly constructed apartment. The solar panels will be installed on a triangular section located at one of the rooftop corners. His engineer designs a 3-D model with corner points proposed to be the points A (1, 2, 3), B (4, 0, 1) and C (2, 1, 5) respectively. He advises that the model will only be effective if the sensor located at (3, 1, -1) lies exactly on the plane surface of the rooftop.

The engineer also proposed that a security camera should be positioned at a point D (0, 0, 10) such that it can directly strike linearly to position A but the inclination of path DA to the solar panel plane should not exceed 15° otherwise some parts will not be captured.

During the system testing, the engineer discovered that distance of coverage by the camera is modelled by $D = (3\sin 2\theta + 4\cos 2\theta - 2)km$. The engineer has target maximum distance of transmission of at least 1.5 km for this will give assurance that all the nearby areas will be covered.

According to the building plan, an elliptical shade will be constructed on the balcony. For better scenery, the shade will be modelled by

$x^2 - 4x + 4y^2 - 8y + 4 = 0$ where x and y are horizontal and vertical distances in tens of metres respectively. A support pillar will be erected at the centre of the shade. The support pillar will be metallic if the distance from the centre to the boundary of the major axis is less than 25m. The engineer also cautioned Mr. Patu that for the shade to create a better view, the eccentricity should not exceed 0.8.

TASK;

Using mathematical evidence and illustrations, help Mr. Patu to;

(a) Establish whether the sensor lies on the surface of the rooftop.
(b) Make a decision on camera installation.
(c) Determine whether the Maximum target distance will be achieved.
(d) Locate the position of the centre of the proposed shade and make the corresponding decisions.

ITEM 2.

A space mining company is planning to send a robotic ship to mine a new asteroid. Before the ship can take off, the team must solve four major problems involving power, space, resources and production goals.

The main drill runs on an advanced twin core battery. Its power output in megawatts after t weeks is modelled by $E(t) = 4^t - 54(2^t) + 350$. The drill will stop working safely if the power output drops to exactly 150 megawatts. The model will be considered void if the power drop occurs in less than three weeks.

The volume of the cave changes depending on the digging depth $x(in\ metres)$ determined using the model $V(x) = x^3 - 12x^2 + 45x - 50$. If the volume becomes zero, the cave will

instantly collapse. The team wishes to know which digging depths will cause a dangerous collapse. The company intends to expand its insurance policy if the cumulative depths of these danger points exceeds 7.5m.

The main station machinery needs three gases for optimal operations; oxygen, nitrogen, and argon. The team has three types of pre mixed gas cylinders A, B and C. Cylinder A is 20%oxygen, 50% nitrogen and 30% argon. Cylinder B is 40% oxygen, 40% nitrogen and 20% argon. Cylinder C is 10% oxygen, 70% nitrogen and 20% argon. The machinery needs an exact total mix of 250 litres of oxygen, 490 litres of nitrogen and 260 litres of argon. There is a standby cylinder with percentage compositions of oxygen, nitrogen and argon as 25%, 40% and 35% respectively. The director of operations advised that the fourth cylinder would only be used for emergency operations if it holds the cumulative volume of all gases that exceeds 350 litres.

To make reasonable profit, two different drilling teams must collect a combined total of at least 15000 tonnes of ore within an 8-week trial. Team 1 mined 500 tonnes in week 1 and increase their production by 50 tonnes every week. Team 2 used a fast growth method where they mined 100 tonnes in week 1 and doubles their production every week. The Management wishes to know if the total amount mined by both teams by the end of week 8 will meet the 15000 tonne target.

Task.

Analyze the mathematical models and justify whether the mining mission is safe and profitable enough to launch.

ITEM 3

Residents in a certain village have been facing challenges of uncoordinated power bills. Mr. Mahato the village chairperson visited the main operations office where the manager gave him some explanations and assured him of getting better services effective the following month.

He was assured that the cost of electricity in ten thousand used per month was computed by a factor N. This factor was derived by subtracting half the number of units used from one. The costs were found by raising this factor to the power of five. Mr. Mahato needed to establish the total units consumed when this factor was equal to 0.875. He was advised to use both the accurate method and the approximate method by first expanding the cost expression and compare the two results. There would be a fair consideration for the households if the resulting percentage error from the exact and approximate findings does not exceed 1.9 %.

In times of power blackout, he has solar panels with an AC converter system whose primary voltage input is regulated by a test signal,

$z = x + iy$. The signal follows a path modelled by $\frac{z-6i}{z+8}$ with two voltage components but secondary voltage component considers only the real part. He can receive maximum lighting from this source if the secondary voltage component follows a linear path with non-negative slope.

Mr. Mahato plans to upgrade his lighting system and purchase a standby generator. He has a budget of sh.15 million and intends to gather this money by the end of December 2026. He was advised by his eldest son to keep depositing sh. 1,000,000 at the beginning of each month in the a Savings and Credit Cooperative Society starting with March 2026 up to the end of December since the SACCO can offer 2.5% compound interest per month. He will not make any withdraws before the expiry of the stated period and will only accept the son's advice if he can achieve his targeted budget amount in the stated saving period

TASK;

As a student of Advanced Level Mathematics, help Mr. Mahato to;

(a) Establish whether there is fair consideration for the households.
(b) Determine whether it will be possible to receive maximum lighting.
(c) Make a decision on following or rejecting his son's advice.

ITEM 4

The students of a certain school have continuously raised concerns over lack of a proper balanced diet. To address the issue, the school Board of Governors decided to implement a fish farming project at school.

Officials from the ministry of Agriculture sent a team to the school which advised them to dig a circular pond of depth 2.0 meters. After the commencement of the project, the head teacher was tasked to make some important decisions.

On a particular day, the pond was being drained for cleaning and then refilled. The rate of change of volume, Vm^3, t hours after draining had started was noticed to be $0.5\frac{dv}{dt}\frac{v+10}{(t+2)(t^2+4)}$; $t \geq 0$ Initially the pond contained $15m^3$ of water. The Board of governors wished to increase the number of fish if after 2 hours the volume was more than $21\ m^3$.

Additionally, the Head teacher noticed that an error of 0.03m had been made in measuring the radius of the pond which was supposed to be 3.0m. He intends to engage technicians from the National Fisheries Research Centre if this error in radius causes a corresponding error in volume greater than 2.5%.

Task;

(a) Help the Board of governors to determine whether they should increase the number of fish after two hours.
(b) Was there need to engage the technicians from the National Fisheries Research Centre

ITEM 5

a certain sub-county in western Uganda, Joseph, a goat's farmer has been facing challenges of theft on his farm.

He wishes to fence his rectangular farm of area $7200m^2$. One side of the garden is a straight river which provides water to the animals and this side will remain unfenced. He plans to minimize his expenses on length of barbed wire needed to enclose the land. He can only achieve this if the smallest possible distance enclosed by the fence is atleast 200 m.

A study program conducted on the farm by a university student revealed that the rate at which the goats multiply on the farm is directly proportional to the number of goats present at that time. Joseph told him that he started with 40 goats and after 1 year, he had 75 goats. He also expressed his desire to extend his farm if the number of goats after 4 years exceeds 400.

A goat was tied with a rope on a tree trunk and when it was released to feed, the rate of change of the length of the rope was 2 ms^{-1}.The farmer intends to keep tethering the goats on some days if the corresponding rate of change in the grazing area for a 2.5m rope exceeds $8\pi m^2 s^{-1}$.

TASK;

Using mathematical justifications, help Joseph to make informed decisions for better performance on his farm.

PRINCIPAL MATHS ONE SET 2

SECTION A

Item 1

A telecom company is expanding its infrastructure in your community. To ensure a successful rollout, you must evaluate the technical feasibility and regulatory compliance in three interconnected projects.

To maintain network stability, the company policy dictates a dynamic load-balancing intervention. If the peak active subscriber count on a single sector exceeds 20,000 concurrent users, management must trigger the "Off-Peak Data Bonus" campaign. This campaign provides data incentives exclusively between 12:00 AM and 5:00 AM to encourage heavy-bandwidth users to shift their usage away from peak operational hours and if the peak hour lies in this range, then the data bonus shifted to another time interval. The total number of active subscribers (in thousands) over a standard 24-hour cycle is modelled by the wave function S(t), where t represents the number of hours past midnight

$(0 \leq t \leq 24)$:

$$S(t) = 13 + 6sin\frac{\pi t}{12} - 6cos\frac{\pi t}{12}$$

The telecom mast is to be installed at the centre of the path of a telecom wave which moves such that its distance from (2,0) is 3 times its distance from (0,4) with all coordinates being in kilometres. There is an existing mast at (20,2) and the company's policy before installation of

new masts requires that the distance from the existing mast to the new mast must be greater than 10km.

There is an existing road modelled as a straight line:

$\frac{x+2}{2} = \frac{y-2}{-1} = \frac{z+1}{-2}$, With x, y and z being in kilometres.

The company has identified a location A (2,1, -5) where it intends to put a support centre for its subscribers on condition that the shortest distance from the road to A is not more than 1 km to ensure easy access for walk-in customers.

Task:

Using your knowledge of geometry, help the telecom company to make valid decisions on their planned expansions.

SECTION B (Part I)

Item 2

Peter is a large-scale farmer planning to start dealing in selling coffee seedlings. Market research shows that the Profit P in tens of thousands of Ugandan shillings is modelled by

$$P(n) = an^2 + bn + c$$

Where n is hundreds of seedlings sold per week.

His friend Hassan already dealing in this business has records of his sales for three different weeks as

$$P(3) = 30,\ P(4) = 56\ and\ P(5) = 78$$

Peter will join this business only if the expected weekly maximum profit made is greater than UGX 1,200,000 and this should be obtained from selling less than 1100 seedlings. He plans to do marketing using his you tube channel with daily live streaming from 6:00pm to 11:00pm. Therefore, he needs a phone whose battery can sustain his live streaming program. However, the energy of his Battery drains according to:

$$E(t) = \frac{4^{(8-t)} \cdot 5^{(16-2t)}}{10^t}$$

where $E(t)$ is the percentage of the battery left after t hours of use. His Phone shuts down when the battery is 10%. He wishes to know if the battery will manage his program or he needs another phone.

His bed for the seedlings is expected to be located at $w = (2 - 12i)$. He plans to open three sales outlets whose expected locations are defined by $Z^3 + 1000i = 0$. However, these outlets must be at a certain distance from the bed location defined by $|w - z| \geq 10\ km$. He doesn't know whether these locations will satisfy his business plan.

Task:

Using Algebra, analyze the information given and give Peter the necessary advice.

Item 3

A juice manufacturing company is experiencing financial distress due to Incorrect expiry date labelling, Weak production and sales planning and Inefficient wage bill policy. Management has set up a 3-part recovery plan.

Anew juice batch is to be launched on 1st August 2026 with initial percentage effectiveness of 98% and sugar concentration constant of 1.2.

The percentage effectiveness P over a time period T in months is related by:

$$\log_3 P + (n-1)\log_3(T+8) = \log_3 P_0$$

where n is the sugar concentration constant in the juice and P_0 is the initial percentage effectiveness in the market. Juice products are considered to be expired when the percentage effectiveness runs below 49%. The committee wants to know the expiry date and has decided that if the juice expires before 1st may 2027 then production should stop immediately.

There are two sales growth strategies for a period of six months all starting with the first month sales of 8000 cartons.

Option A - Add 2000 more cartons each month.

Option B - increase sales by 20% each month.

The committee wants to know which option gives higher total sales by the 6th month and by which margin.

The remainder when the total wage bill $W(x)$ is divided by $(x-500)$ is 160,000 and when it is divided by $(x+400)$, the remainder is 70,000, where x is the current basic fee. The committee plans to increase the basic fee from 20,000 to 25,000 if the remainder when $W(x)$ is divided by $x^2-100x-200{,}000$ is less than 2,650,000 basing on the current basic fee.

Committee is not sure if the wage increase should be approved.

Task:

Use your knowledge of algebra to analyze the information above and help the committee to make decisions.

Part 2

Item 4

Your community has been affected by a contagious disease. The health officers have decided to set up a quarantine Centre that can only handle 250 cases.

It has been observed that the disease occurs at a rate proportional to the product of the number of people infected and those not yet infected. Initially, out of a total population of 1000, 50 are infected and the disease is spreading at a rate of 10 new cases per day. The community members claim that by the 9th day the Centre will be full beyond its capacity and the government should intervene as quickly as possible by taking away the excess number of patients to the nearby hospital.

The district health officers are planning to construct a rectangular fence across the quarantine Centre with one gate measuring 12 m in length. There is 1000m of fencing available.

The construction team wants to know the dimensions of the field that will yield a maximum area that can be fenced off before starting the installation.

However, the community is constrained with funds and they can manage to fence a total area of $64020m^2$ only.

The community members are not sure if the fences can be constructed given the available funds or they collect funds from each family.

Task

(a) Determine whether the community's claim is correct and advise the government on the number of patients to be transferred by the 9th day.

(b) Advise the construction team and the community about the fencing.

Item 5

A farmer is planning ahead for his flock. He has 3 main challenges: goat numbers, water supply, and shelter. He wants mathematical advice before making expensive decisions. On 1st April 2025, the farmer had 40 goats. On 1st October 2025, the flock had grown to 60 goats. It is observed that the number of goats grows at a rate proportional to the number of goats N, present at a time t months.

For sustainable grazing, he intends to keep a maximum of 135 goats. He needs to know when he will reach that number so he can plan to sell excess goats.

He uses 900 litres of water per day. He needs a reservoir that can store enough water for at least 18 days.

He has one aluminium sheet measuring 8 m by 5 m. A welder can make an open rectangular tank by cutting equal squares from each corner and folding up the sides.

If the largest tank from this sheet cannot store 18 days of water, he will dig a dam instead. He has 20 iron sheets, each of area 2.7m². He wants to roof a shelter whose roof surface has the shape of the region bounded by the curve $y = 60x - 10x^2$ and the line $y = 10x + 40$ All dimensions are in metres.

He needs to know if his iron sheets are enough, or if he should buy more before calling a roofer.

Task

Using your knowledge of calculus, advise the farmer on:

(a) When to plan sales,

(b) Whether he should build the tank or dig a dam,

(c) The number iron sheets.

END

NUMERICAL METHODS

ITEM 1

An agricultural engineer, Mr. Kato, is designing an irrigation pipeline to supply water to a farm. The irrigation pipeline follows the path $y = lnx - 4 + x$ where x and y are horizontal and vertical distances respectively measured in kilometers

The main water-supply pipe is represented by the straight-line $y = 0$. Mr. Kato needs to plan for the connection by estimating graphically the point where the irrigation pipeline meets the main water-supply pipe between $x = 2.5km$ and $x = 3.5km$ from the land mark stone P of the farm. To determine the connection point more accurately, Mr. Kato uses a suitable iterative method and records it to three decimal places. The farm management will approve the connection only if the accurate location lies within 2.95km from the land mark stone P.

Additionally, the area enclosed between the irrigation pipeline and the main water-supply pipe over the interval from $x = 3km$ to $x = 3.5km$ is to be used to store irrigation equipment and other farm materials. Mr. Kato wants to first estimate this area using a suitable approximation method with five ordinates. He will also determine the required area using an exact method, and if the percentage error in estimating this area is less than 1% the construction of the storage room will be approved.

Task

Help Mr. Kato obtain the mathematical information required to make informed decisions about the approval of the irrigation pipeline and the storage room.

ITEM 2

A construction company involved in land development and property investment has recently purchased a rectangular plot of land for future residential and commercial development. The plot owner says, his plot measures 104.65 m by 26.2 m, with both dimensions correct to the nearest decimal place.

The company's technical team is required to determine the possible range of the actual area of the plot, and to assess whether the absolute error in the estimated area does not exceed 5 m², in order to approve payment.

The company is also reviewing its annual financial performance to guide future investment decisions and tax planning.

The financial estimates are as follows:

- Estimated annual profit = 25 million UGX, with a margin of error of ±3%
- Estimated annual loss = 4 million UGX, with a margin of error of ±2%
- The Uganda Revenue Authority has assessed an annual income tax of 3.5 million UGX

The management wants to determine the possible percentage range of gross income that goes to tax.

To improve employee welfare and long-term financial security, the company introduces a staff savings scheme. Employees can invest a portion of their salary, which earns compound interest annually at a rate of $r\%$.

The savings model is given by:

- Initial investment: P
- Annual interest rate: $r\%$
- Duration: $n = 5$years
- Amount after n years: A_n

The company uses the compound interest model to track growth and encourage participation in the scheme.

To automate the savings scheme, the company requires a clear flowchart-style algorithm that:

1. Reads the initial amount P_1, previous amount A_1, and interest rate r
2. Iterates the compound interest process for n = 5 years
3. Computes and prints the accumulated amount A_n

The correctness of the algorithm is verified using a test case for a period of 5 years:

- $P_1 = 100{,}000$UGX
- $r = 15\%$

Tasks.

As a student of math perform a mathematical analysis to;

(a) Assess whether the plot will be bought

(b) Determine the percentage range of gross income that goes to tax

(c) Design a flow chart algorithm and verify its correctness.

DATA ANALYSIS & PROBABILITY

ITEM 3

The Ministry of Health has organized a health awareness day at a district Health Centre IV to promote better health services and improve community participation. During previous health awareness activities, patients frequently complained about delays in accessing health services, which were partly attributed to the late arrival of health worker.

In order to improve service delivery and ensure that health workers arrive on time, the ministry collected data on the time taken by health workers to travel to the hospital during the previous month's health awareness day. The recorded travel times for 400 health workers are shown in the table below

Time (minutes)	<5	<10	<15	<20	<25	<30	<35	<40	<45
Number of health workers	17	45	78	143	280	340	375	395	400

The ministry contacted a statistician who advised them to determine the dominant time taken by health workers to travel to hospital, he also recommended for construction of residential houses for the health workers if the standard deviation of time taken by health workers to travel to the hospital is more than 15 minutes

When the data was analysed by the statistician, it was found out that it was approximately normally distributed. He advised the ministry that there was need to know the 95% confidence interval around the average score, and if it lies between 35 and 45, then the starting time for the next health awareness day will be readjusted.

Additionally, the statistician observed that health workers travelled to the health centre using personal cars (C), buses (B), and taxis (T), with proportions of 40%, 50%, and 10%, respectively. The probabilities of arriving late among health workers using personal cars,

buses, and taxis were 0.09, 0.06, and 0.20, respectively. The statistician proposed that the ministry should introduce a special van service to transport health workers from their residences if the probability that a health worker used a taxi, given that the worker arrived late, is greater than 0.2.

Task

As a student of mathematics,

(a) Help the ministry to determine the dominant time and ascertain if there will be construction of residential houses for the health workers.

(b) Assess whether the starting time for the next health awareness day will be readjusted.

(c) Determine whether a special van will be put in place to pick all health workers from their residences.

ITEM 4

In Kampala Northern By-pass, road accidents are increasing due to dangerous practices such as reckless crossing by pedestrians and over-speeding by motorists. Consequently, traffic-control measures, including traffic lights and road humps, have been introduced on busy roads.

At a particular point along the by-pass, the speeds of cars are normally distributed. Police investigations established that: 95% cars travelled at a speed less than 85 kmh^{-1} and 10% travelled at a speed less than 55 kmh^{-1}. According to traffic regulations, any car travelling at a speed exceeding 65 km h^{-1} shall be impounded.

Further, the traffic police sampled 100 cars along the by-pass to investigate their average speed. The Officer in Charge of Traffic directed that new road humps would be constructed if the proportion of the cars travelling at an average speed greater than 70 km h^{-1} exceeds 0.45.

In addition, the police inspected 300 cars that were in dangerous mechanical condition (DMC). The ages of these cars, X, measured in tens of years, were found to follow the continuous probability distribution given by

$$f(x) = \begin{cases} k, & 0 \le x \le 1.5 \\ \frac{k}{2}(2-x), & 1.5 \le x \le 2 \\ 0, & else\ where \end{cases}$$

Cars older than 1.6 tens years will be taken to the police station for further forensic analysis.

Tasks

As a senior six student if consulted, come up with a clear report on the following backed by calculations:

(a) The percentage of cars that shall be impounded

(b) Whether new road humps will be constructed or not.

(c) The number of DMC cars that were taken for forensic analysis

ITEM 5

During a national Presidential and Member of Parliament (MP) campaigns of a certain country, a research team from election observers is studying number of numbers (in thousands) using two different areas.

- Urban area contains 4 respondents supporting MP Candidate White and 5 respondents supporting MP Candidate Pink
- Rural area contains 4 respondents supporting MP Candidate White and 3 respondents supporting MP Candidate Pink

Due to population differences, urban area is twice as likely to be selected as rural area.

An election observer randomly selects one area and then randomly interviews two respondents, each respondent being interviewed only once.

The research team wishes to make an assessment on the;

- probability of the two selected respondents, both support the same MP candidate.
- probability of the two respondents selected being from an urban area, given that they both support the same MP candidate.
- Probability distribution of the number of correspondents who support candidate Pink and Expectation of the distribution.

At the same time, the campaign manager of Pink was moving around urban areas to determine the number of people who support MP candidate Pink and Presidential candidate Pink. The findings were summarized in the table below.

MP candidate Pink (x)	78	66	73	75	84	66	89	84	67	77
Presidential candidate Pink (y)	81	68	81	75	80	67	85	83	66	78

Campaign team of candidate Pink wishes to;

- Represent the information graphically and draw a line of best fit to ascertain the trend of the campaigns.
- Estimate the number of peoples who support Presidential candidate Pink, if MP candidate pink has 80 supporters.
- Change their campaign strategy, if the ranks correlation coefficient is less than 0.8

Tasks

As mathematic student help the;

(a) Research team make an assessment.

(b) Campaign team of candidate pink perform an analysis and determine if they should maintain their campaign strategy.

ITEM 6

A farmer applied a specific fertilizer during the planting season to improve crop growth and productivity. The table below shows the number of days taken for different groups of plants (in hundreds) to reach maturity and be ready for harvest.

Number of days	30	35	40	50	60	70	80
Number of plants	2	7	9	10	8	6	3

To assess the efficiency of the fertilizer, the farmer would like to determine its effectiveness using the equation;

$$Effeciency = \frac{Standard\ deviation\ of\ days}{Median\ number\ of\ days}$$

If the efficiency is less than 0.4, he will increase the amount of fertilizer used on the farm.

It is also observed that the number of days to maturity is approximately normally distributed. The farmer further wishes to determine the probability that a plant will mature

less than 60 days. If this probability exceeds 0.5, the farmer will place more orders for the fertilizer

In addition, two farmers, A and B, who used the same fertilizer, supplied produce to the market. The probability that a randomly selected item comes from farmer A is 0.2, and from farmer B is 0.8. It is known that 5% of produce from farmer A and 8% from farmer B are defective. If a single item is randomly selected and found to be defective, the vendor wants to determine the probability that it originated from farmer A. If this probability is greater than 0.3, the vendor will decide not to place another order from farmer A.

Tasks

(a) As a mathematics student, justify the efficiency of the fertilizer
(b) Decide whether the farmer should place more orders for the fertilizer
(c) Advice the vendor on whether to place another order from farmer A.

MECHANICS

ITEM 7

With the aim of ensuring that the transport truck arrives at its destination on time, the site engineer, plans the journey from Town A, B and C. A Sino truck carrying construction bricks passes through three town A, B and C which lie on a straight road with AB=8.5 km and AC=28 km. The truck moves along the straight road with a constant acceleration of $\frac{1}{80}\ ms^{-2}$, it also take it 10 minutes to travel from A to B. For the truck to make two trips of bricks, the time taken for the truck to move from town B to town C should be less than 20 minutes.

At construction site, bricks are raised up the building site by means of a vertical rope tied to one end of light bucket. At every trip, a pack of bricks of mass 100kg is to be hoisted up the side of a building at an acceleration of $0.25ms^{-2}$. The process of raising the bricks will be considered safe and free from danger if the tension in the string attached to the bucket is more than 1050N.

Additionally, after offloading the bricks, the bucket is returned to the ground by throwing it from the edge of the vertical building 52.5m high with a velocity of $2\ ms^{-1}$ at an angle $\tan^{-1}\frac{7}{24}$ above the horizontal. The bucket hits ground at a point from the base of the building. The site engineer wants to ascertain the safety of the site workers who are standing 10 m from the base of the building.

Task

Help the engineer to obtain the mathematical information required to make informed decisions.

ITEM 8

The airport traffic control tower is monitoring the safety of two aircrafts **A** and **B** with **A** landing and **B** taking off from the airport. At 7:30am, the airport traffic control officer observed the aircrafts A and B flying at the same altitude with velocities $200ms^{-1}$ in the direction $N30^0E$ and $300ms^{-1}$ in the direction $N50^0W$ respectively. Initially A and B are 2km apart with B on a bearing $S70^0E$ from A. According to civil aviation regulations, for any two aircrafts to be safe in airspace, the closet distance between them should be more

than 1.5km otherwise they are to be advised to change their altitude to be safe. The airport traffic control officer needs to determine whether the two aircrafts are safe to continue with their flights or change their altitude.

Aircraft **A** of mass 500 tonnes lands on a horizontal level runway at a speed of 40 m/s. After touchdown, its engines provide a forward driving force of 20,000 N, with the coefficient of frictional of 0.14 between the tyres and the tarmac. For the aircraft to land safely, it should come to rest within 1,200m on the runway after touchdown. The pilot wants to know weather the aircraft will slow down and land safely on runway.

After the aircraft **A** came to a complete stop on the tarmac, mobile passenger boarding stairs were positioned beside the aircraft to allow passengers to exit the aircraft safely. The stairs have an inclined structure supported by an isosceles triangular frame ABC right angled at A with AB= 2m and all its sides acted upon forces to keep the stairs stable. Forces of 800N, $400\sqrt{2}$N and 600N act along BA, BC and CA respectively. The airport management want to whether the frame will be stable to with stand the passenger's weight. According to regulations, the frame is considered stable if the resultant force in the frame is more than 3000N.

Task

As a mathematician, analyse the data and provide a methodology and logical judgements to the airport management team.

ITEM 9

A house owner wishes to install rectangular metal signboard $ABCD$ on the roof of his building. He hires an engineer to install the signboard; he finds out that signboard has dimensions $AB = 0.5$ m and $AD = 1.5$ m. To assess the structural stability of the signboard, forces of magnitudes 15 N, 12 N, 16 N, 10 N, and 20 N act along the sides AB, CB, CD, AD and the diagonal AC respectively. Engineering guidelines require that the resultant force acting on the signboard should not be less than 20 N for it to remain securely fixed.

The house owner realises that during a severe windstorm, the signboard became detached and started sliding down a rough section of the roof. The roof is inclined at 20°to the horizontal and has a coefficient of friction of 0.3. Starting from rest, the signboard slide for 2 seconds before reaching the edge of the roof, which is 3 m above the ground. The engineer noticed that the signboard left the roof and moved through the air at an angle of 30° below the horizontal. The engineer is worried about the signboard falling on the flowers on the ground planted 1m from the wall.

Task

As a mathematics student:

(a) Advise the engineer whether the signboard is securely fixed based on the forces acting on it.
(b) Determine whether the signboard will fall on the flowers

ITEM 10

An agricultural cooperative in Kamuli District has harvested a large quantity of fresh produce from farms in the surrounding areas. The cooperative has been contracted to

deliver the produce to a storage facility on an island in Lake Kyoga. The delivery must be completed within a strict schedule because the produce is highly perishable and is required for a community food-supply programme.

Early in the morning, a lorry driver departs from a collection centre in Kamuli carrying the harvested produce. The driver has been informed that the ferry connecting the mainland to the island is scheduled to depart from the landing site in 10 minutes' time. Missing the ferry would delay the delivery by several hours and increase the risk of spoilage of the produce.

The lorry is initially 20 km from the ferry landing site. It has a mass of 3.5 tonnes and moves along a rough horizontal road. The engine exerts a constant driving force of 1.5 kN, while resistive forces due to friction and air resistance amount to 0.25 N per kilogram of the lorry's mass.

The agricultural cooperative manager wishes to determine whether the lorry can reach the landing site before the ferry departs.

Upon arrival at the landing site, the lorry is loaded onto the ferry for transportation to the island. The ferry captain must navigate directly to an island located due west of the landing site. However, the captain knows that the lake has a current flowing at 3 km h^{-1} in the direction S28°E. The ferry is capable of moving at 10 km h^{-1} in still water.

To ensure that the ferry reaches the island without drifting off course, the captain must determine the direction in which the ferry should be steered.

After reaching the island, the produce is transported to a cold-storage facility where it is unloaded. Some lightweight produce crates are discharged from the back of the lorry by allowing them to leave the loading platform horizontally at a speed of 10 m s^{-1} from a height of 2 m above the ground.

To prevent damage to the produce, safety guidelines require that the crates should not strike the ground with a speed exceeding 15 m/s. The storage facility supervisor therefore wishes to assess whether the unloading procedure satisfies the safety requirements.

Tasks

As a mathematics student, help the:

(a) Agricultural cooperative manager to determine whether the lorry can reach the ferry landing site before the ferry departs.

(b) Ferry captain to determine the direction in which the ferry should be steered.

(c) Storage facility supervisor to assess whether the unloading procedure is safe.

END